

A prime distinguishes the actual theta lift from its synchronization

$$\mathcal{A}_L \longrightarrow L[C_2] \longrightarrow \mathcal{P}(C_2), \quad (f, v) \longmapsto (v_0, v_1)$$

Full lattice product: $(a, b)(c, d) = (a \wedge c \vee b \wedge d, a \wedge d \vee b \wedge c)$

$\chi: L \rightarrow \mathbb{B}$ is a stated receiver map; all labels remain in the preceding full-L construction.

Neither parity $\emptyset$ $(0, 0)$ in the prime	Even only $\{0\}$ $(1, 0)$ outside the prime	Odd only $\{1\}$ $(0, 1)$ outside the prime	Both parities $\{0, 1\}$ $(1, 1)$ in the prime
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$$\mathfrak{p}_{\text{mix}} = \{\emptyset, C_2\}, \quad \{1\} \cdot \{1\} = \{0\}, \quad C_2 \cdot A = C_2 \quad (A \neq \emptyset)$$

This is an ordinary semiring prime ideal. It is not a downward or subtractive ideal.

The two original lifts have the same function and different prime membership

$$\mathbf{H}_t = (H_t, (1_L, 0_L, 1_L, 0_L, \dots)) \longmapsto \{0\} \notin \mathfrak{p}_{\text{mix}}$$

$$J\mathbf{H}_t = (H_t, (1_L, 1_L, 1_L, 1_L, \dots)) \longmapsto C_2 \in \mathfrak{p}_{\text{mix}}$$

$$\mathbf{H}_t^{\star t^k} = (T_t(H_0^k), (1_L, 0_L, 1_L, 0_L, \dots)) \quad (k \geq 1)$$

Every tensor power keeps the same distinction. Inverting J removes this prime.

Proofs: SHP17–28, HTK24–32 and the complete HFP prime-spectrum calculation.

The diagram shows the exact four-element parity receiver, not an exhaustive picture of the full spectrum.

Kernel: Brad Rodgers & Terence Tao, arXiv:1801.05914v5. Full carrier: The Clankers, v11, def:lattice-split.